

The Sign of the Binder: V5’s Attraction from Substrate Coherence

The last open structural piece of ED’s cross-chain kernel, forced from the coherence content by the substrate’s own gauge law and build-verified; a companion to Knots, Not Crystals and The Relational Tick

Allen Proxmire

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Preamble: what this paper does NOT claim

Written first, per house discipline; the abstract is reconciled against it.

1. **The 13 primitives are not derived** (Paper_087).
2. **V5’s existence is not derived.** V5 stays a P10-licensed kernel rule-type posit, on exactly the same footing as V1. Paper_090 §1.1 is explicit: P10 “opens room for” cross-chain kernels, it does not force any. This paper derives V5’s *sign*, not its existence.
3. **V5’s reach and memory values are not derived.** The scales ℓ_{V5} and τ_{V5} are inherited from value-layer content, as Paper_090 §3.3 already states. This paper adds a mechanism for the *finiteness* of the reach, not its number.
4. **“Derived” here means forced-conditional**, on two commitments that already exist in the corpus and are not invented for this result: P-Bipartite-Mapping (V5’s cross-chain content is bilinear in the two chains’ participation amplitudes, Paper_063 §3) and the reading of V5 as a correlation between like chains. It is not a derivation from nothing.
5. **The build is on a polarity-extended substrate rule.** The certified ρ -dynamics are used verbatim; a P09 phase channel and a P12 coherence coupling are *added* to the certified selection functional. No certified result is altered.
6. **This is not a new empirical prediction.** It supplies the attractive sign that the clock-binding result (The Relational Tick) and the MOND-interference result (Interference Gravity) previously assumed. The observational content of those results is unchanged.
7. **It does not claim the coherence functional is the only coupling that could supply the sign.** It claims that V5’s coupling, as constrained by Paper_090’s own gauge law, *is* this one, and that this one is attractive.

Abstract

V5, ED’s cross-chain correlation kernel, is the program’s highest-leverage keystone: it is inherited in the entanglement, dynamics, black-hole, and cosmology arcs and derived in none of them. Its home paper (Paper_090) already fixes most of its structure from the primitives, its existence as a P10 kernel, its retarded support from P11, and its gauge-covariant phase from P05 and P09, and leaves the *sign* of the coupling envelope open, the one open piece that decides whether V5 can bind at all. That sign must be attractive for V5 to bind nearby chains into a shared local clock, but across the corpus it is assumed.

This paper derives it, in two steps. First (the identification), Paper_090’s own gauge law is a *difference* phase, $K_{V5} \rightarrow e^{i(\alpha_A - \alpha_B)} K_{V5}$, which is the fingerprint of a conjugated amplitude-pair moment $\langle P^A (P^B)^* \rangle$; the joint-amplitude *product* $P^A P^B$ transforms as the *sum* $e^{i(\alpha_A + \alpha_B)}$ and is a different object. The gauge-invariant content of that moment is the P05-connection-dressed phase overlap of the two chains, the coherence content

of the cross-chain superposition, $\sqrt{b_A b_B} \cos(\Delta\pi - \int \mathcal{A})$. So V5's coupling is the coherence content. Second (the build), coupling two clusters of real certified chains by that coherence content, entered into the certified selection functional with P12's own reward-coherence sign and with no hand-put attractive term, reproduces V5's characteristic signature: nearby chains lock to a shared proper-time rate while distant clusters keep their rate gap, and this collapses to a single universal tick only when the reach reaches the separation. Flipping to the anti-coherence sign removes the local lock at every reach. The attractive sign is therefore not a free assumption; it is inherited from P12's stability landscape rewarding coherence. Tier: measured, with the identification gauge-law-forced, conditional on P-Bipartite-Mapping and the like-chain reading. Existence stays a posit; the reach value stays inherited.

1. The keystone and its one open piece

Event Density reads the smooth structures of physics, spacetime, fields, constants, as the coarse-grained pattern made by discrete relational events. Two substrate kernels sit under most of the dynamics: V1, a single-chain vacuum-response kernel, and V5, a cross-chain correlation kernel. V5 carries an unusual load. It is the shared mechanism under the entanglement arc's non-factorizability, the dynamics arc's stress-energy form, the black-hole arc's Hawking-cutoff structure, the soft-matter arc's Maxwell relaxation, and, as the companion paper *The Relational Tick* shows, the binder that assembles nearby chains into a composite object with a single shared proper-time clock. It is postulated in each of these and characterized from the primitives in none. Closing any structural piece of V5 therefore has the widest reach of any single technical move in the program.

Most of V5's structure is in fact already pinned. What is not pinned, until this paper, is the one thing that decides whether V5 can do its job at all: the *sign* of its coupling. A binder must attract. If V5's cross-chain coupling pushed phases apart rather than together, there would be no local clocks, no composite rest frames, no bound matter. The synchronization study behind *The Relational Tick* established that the attractive sign is a *necessity condition*, V5 must be attractive to bind, but it assumed the sign rather than deriving it. This paper removes the assumption.

2. What Paper_090 fixes, and what it leaves open

Paper_090 defines V5 through the cross-chain correlation

$$\langle X^A(u_A, t_A) \cdot Y^B(u_B, t_B) \rangle_{V5} = \int K_{V5}(u_A, t_A; u_B, t_B) \mathcal{Q}_{X,Y}^{AB} d\mu(\text{intermediate}),$$

with the kernel

$$K_{V5}(u_A, t_A; u_B, t_B) = \theta(t_A - t_B) F_{V5} \left(\frac{\sigma(u_A, u_B)}{\ell_{V5}^2}, \frac{t_A - t_B}{\tau_{V5}} \right).$$

Here θ is the Heaviside step, F_{V5} is a bounded, decaying envelope, σ is a substrate Synge-function analog, and $\mathcal{Q}^{AB}$ is the cross-chain participation content the kernel integrates against. Reading the paper's own attributions:

- **Existence** is a P10 posit. P10 licenses kernel rule-types beyond V1; it does not force them. V5 is on the same footing as V1 (§1.1, §3.5).
- **Retarded support**, the $\theta(t_A - t_B)$, is forced by the same P11 commitment-irreversibility argument that fixes V1's retardation (§3.4, §4.6).
- **The gauge phase** comes from P05 and P09: under a $U(1)$ transformation, $K_{V5} \rightarrow e^{i(\alpha(u_A) - \alpha(u_B))} K_{V5}$ (§4.3).
- **The reach and memory scales** ℓ_{V5}, τ_{V5} , and the specific envelope shape of F_{V5} , are inherited value-layer content (§3.3).

The one structural item with no derivation anywhere, and the one that decides whether V5 can bind, is the **sign** of the coupling. F_{V5} is introduced as a general bounded envelope; nothing fixes whether the coupling attracts or repels. (The detailed envelope shape is also open, but it does not bear on the sign and is a separate thread.) The sign is the target.

3. The identification: V5's coupling is the coherence content

The envelope $K_{V5} = \theta F_{V5}$ is real, so it cannot be what carries the gauge phase; a real object cannot transform by $e^{i(\alpha_A - \alpha_B)}$. The phase must live in the participation content Q^{AB} that the kernel multiplies. (Paper_090 writes the gauge law on K_{V5} ; we resolve that looseness the only internally-consistent way, by locating the phase in the content. This is the section's one interpretive step, and it is forced by consistency, not chosen.)

Now use the canonical participation amplitude $P^C = \sqrt{b_C} e^{i\pi_C}$ (P04 sets $|P|^2 = b$, P09 sets the phase π), which transforms under the $U(1)$ as $P^C \rightarrow e^{i\alpha(u_C)} P^C$. Two bilinear objects can be built from the two chains' amplitudes, and they transform differently:

$$\begin{aligned} P^A P^B &\rightarrow e^{i(\alpha_A + \alpha_B)} P^A P^B && \text{(sum phase),} \\ P^A (P^B)^* &\rightarrow e^{i(\alpha_A - \alpha_B)} P^A (P^B)^* && \text{(difference phase).} \end{aligned}$$

Paper_090's gauge law is the **difference** phase. That is a fingerprint. The product $P^A P^B$ is exactly Paper_063's factorizable joint amplitude $\Psi_{KL}^{AB} = P_K^A P_L^B$ (the bipartite *state*), and it carries the *sum* phase. So V5's coupling is **not** the joint-amplitude product. The object that carries the difference phase is the conjugated amplitude-pair moment $P^A (P^B)^*$ (unique up to complex conjugation: $(P^A)^* P^B$ carries $e^{-i(\alpha_A - \alpha_B)}$ and has the same real content), the substrate analog of a two-point correlation or propagator, which is what a correlation kernel between two chains should be. Writing it out,

$$P^A (P^B)^* = \sqrt{b_A b_B} e^{i(\pi_A - \pi_B)}.$$

Its modulus $\sqrt{b_A b_B}$ is manifestly gauge-invariant, and its phase $\Delta\pi = \pi_A - \pi_B$ is gauge-covariant: under the transformation it shifts by $\alpha_A - \alpha_B$. The gauge-invariant phase observable is therefore the P05-connection-dressed difference $\Delta\pi - \int_B^A \mathcal{A}$, in which the P05 connection $\mathcal{A}$ compensates the shift. This is the same P05 connection that transports the phase in the kernel, so P05 enters consistently in both roles. The gauge-invariant coherence content is then

$$\text{Coh}_{AB} = \sqrt{b_A b_B} \cos\left(\Delta\pi - \int_B^A \mathcal{A}\right),$$

which reduces to $\sqrt{b_A b_B} \cos \Delta\pi$ in a fixed gauge or trivial-connection limit, and is the interference part of the superposition, $|P^A + P^B|^2 - b_A - b_B$ up to the factor 2. This is the same functional that *Phase Coherence in ED* operationalizes for P12 and build-verifies as finite-reach, and it is exactly the connection-dressed coherence the build of §4 uses, whose holonomy A_{ij} is the discrete $\int \mathcal{A}$. So **V5's phase-dependent coupling is the coherence content of the two chains' superposition**, modulated by the real reach-and-memory envelope F_{V5} .

This identification is the gauge-consistent one, forced among content bilinear in the two chains' amplitudes, conditional on three commitments, none invented here:

1. **The gauge law** $e^{i(\alpha_A - \alpha_B)}$, stated verbatim in Paper_090 §4.3.
2. **Bilinear cross-chain content**: V5 “acts on pairs of chains” (§3.2), and Paper_063 §3 builds the bipartite content from the two amplitudes ($\Psi_{KL}^{AB} = P_K^A P_L^B + \Delta_{KL}^{AB}$, the product piece being P-Bipartite-Mapping). Among content bilinear in the two amplitudes, the conjugated-pair moment is the unique object with the difference-phase transformation.

3. **The like-chain correlation reading.** A formal alternative also produces a difference phase: two *opposite-charge* chains multiplied as a product, $P^A P^B$ with $P^B \rightarrow e^{-i\alpha_B} P^B$, giving $\cos(\pi_A + \pi_B)$. That alternative is physically excluded, because $\cos(\pi_A + \pi_B)$ is maximal at anti-alignment $\pi_A = -\pi_B$ and would make chains repel, contradicting V5’s binding role. The like-chain correlation ($\cos \Delta\pi$, maximal at alignment) is selected by V5’s function, not assumed at random.

This is a far cleaner dependency than the program’s earlier attempt to characterize V5 through Tsirelson’s theorem, which was retracted because it needed measurement operators, a commuting tensor split, and a Born bridge that Paper_063 never builds. The present argument uses only what Paper_063 actually builds (the bilinear joint content) plus Paper_090’s gauge law and the standard correlation reading. No operator algebra is invented.

4. The build: the attractive sign emerges from P12’s coherence

The identification says V5’s coupling is the coherence content. The attractive sign then has a candidate source that is not V5-specific: P12’s stability landscape is $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$, in which the coherence term enters with a *positive* coefficient by the definition of P12. The positive coefficient attaches to whatever functional plays the coherence role; that the phase-coherence content of §3 is that role, rather than the certified density-based $\text{Coh} = -(\rho - \rho^*)^2$, is the operationalization established in *Phase Coherence in ED*. If V5’s coupling is that coherence content, its sign is P12’s sign. We test this directly on the certified substrate, taking care to put in no attraction by hand.

Setup. Two clusters of four chains each run on parallel lanes. Each chain is a real certified 1D chain: the advance-versus-dwell decision at each step is the certified Σ (`compute_sigma`) verbatim, and natural rates come from the established memory mechanism, more carried memory means more dwelling means a slower rate. The two clusters run at different mean rates with a genuine within-cluster spread, a time-dilation setup: two composites at different proper-time rates, each with real internal spread that would drift apart uncoupled. Each front carries a $U(1)$ gauge phase ϕ that advances with its forward proper-time progress, a proxy in which ϕ tracks accumulated forward commitments (P09; Paper_090 §4.3; flagged as a modeling choice). The only cross-chain coupling is P12’s coherence gradient, entered into the certified selection with P12’s own sign:

$$\text{sig}_{\text{advance}} += k_{c5} \sum_j w_{ij} \sin(\phi_j^{\text{ret}} + A_{ij} - \phi_i),$$

where $w_{ij} = e^{-|\Delta y_{ij}|/\ell_{V5}}$ is the transverse reach weight (times a longitudinal window), A_{ij} is a quenched P05 connection carrying substrate disorder, and ϕ_j^{ret} is the retarded (previous-step) neighbor phase (Paper_090 §4.6, P11). The sin is the gradient of cos-coherence with respect to ϕ_i : advancing moves a front up the coherence gradient, toward its neighbors’ phase. The sign $k_{c5} > 0$ is P12’s reward-coherence sign. **No attractive term is put in.** Whether $+k_{c5}$ synchronizes is read off.

A note on method, kept honest. The first construction compared each choice’s coherence against the neighbors’ frozen last-step phase. At the aligned start that rewards phase *stasis*, holding still, rather than rate-matching, and the result was sign-independent: a null. The correct entry of “reward coherence” into a rate decision is the coherence *gradient* above, which is the exact derivative of the coherence and points a phase-behind front toward advancing. The reported runs use the gradient form. We record the null and the fix because ruling out the stasis artifact is part of trusting the result.

Results (three seeds, 2500 steps, cluster separation $D = 20$). The uncoupled baseline has within-cluster rate spread 0.084 and cross-cluster gap 0.243.

condition	reach ℓ_{V5}	within-spread (ratio to baseline)	cross-gap	reading
P12 sign (+)	2 to 30 ($< D$)	0.022 to 0.025 (0.27 to 0.30)	0.21 to 0.24 (preserved)	local lock, time dilation intact

condition	reach ℓ_{V5}	within-spread (ratio to baseline)	cross-gap	reading
P12 sign (+)	100 ($\geq D$)	0.027 (0.32)	0.036 (collapsed)	universal tick
sign-flipped (−)	all	0.062 to 0.065 (0.74 to 0.78)	~ 0.21	no lock at any reach
(+), quenched disorder	2 to 30	0.023 to 0.025 (0.27 to 0.30)	preserved	local lock, disorder-robust

Three readings. First, with P12’s sign and a finite reach smaller than the separation, the within-cluster spread collapses by about 73 percent: nearby chains lock to a shared rate, a local proper time, while the two clusters keep their different rates, time dilation preserved. When the reach reaches the separation, everything locks and the gap collapses to a single universal tick, the wrong regime, confirming that the *finiteness* of the reach is load-bearing for time dilation. This is V5’s characterized signature, emerging from $+k_{c5}$ alone. Second, flipping to the anti-coherence sign removes the local lock at every reach: the lock is specifically P12’s positive coherence sign, not the coupling machinery or the reach structure. Third, adding the substrate’s own quenched disorder to the P05 connection does not spoil the local lock; if anything it protects against the universal-tick collapse at long reach.

The small sign-independent within-spread reduction (ratio around 0.72) that appears in the flipped and short-reach runs is a uniform perturbation from the coupling breaking some ties. It is not synchronization: it has no reach dependence and does not preserve the gap structure. The genuine lock (ratio to 0.27 with the gap preserved and a sharp transition at $\ell_{V5} \approx D$) is only present for the positive sign at finite reach.

5. What this closes, and what it does not

Combining the two results: the build shows V5’s attractive local-synchronization signature emerges from P12’s positive coherence sign with a passing sign-flip control, and the identification shows the coupling that carries that sign is forced by Paper_090’s own gauge law to be the coherence content. So V5’s attractive sign is derived from P12’s reward-coherence structure. The ledger:

piece of V5	before	after this paper
existence	P10 posit	P10 posit (unchanged, as for V1)
retarded support	P11-forced (090)	P11-forced
gauge phase	P05 + P09 (090)	P05 + P09
attractive sign	assumed	derived, forced-conditional, from P12 coherence
coupling = coherence content	not stated	derived-conditional, via the gauge law (G1)
reach, memory values	inherited	inherited
envelope shape F_{V5}	open	open (separate thread)

The tier is **measured** for the emergence of the sign on the certified substrate, with the identification behind it **gauge-law-forced**, conditional on P-Bipartite-Mapping and the like-chain correlation reading. What is explicitly not closed: V5’s existence stays a posit, exactly as V1’s does, because P10 licenses kernels without forcing them; the reach and memory values stay inherited; the temporal envelope shape stays open. The honest residuals are four: Paper_090’s §3.1 loosely attributes the gauge phase to the real envelope, which we resolved by consistency; the bilinearity rests on P-Bipartite-Mapping, a postulate rather than a primitive theorem, though a pre-existing one; the identification is for the amplitude-level correlation, taking the correlated observables X, Y as the participation amplitudes themselves, whereas Paper_090’s content $Q_{X,Y}^{AB}$ for a general observable pair is dressed (the clean coherence reading is the amplitude-amplitude case, which is the one relevant to the phase and the binding); and the intermediate integral $d\mu$ could dress the leading two-chain term in the multi-intermediate case.

6. Relation to the companion results

This single object closes assumed signs in three companion papers at once. *The Relational Tick* needs V5 to be attractive to bind chains into local clocks; that sign is now derived, and this paper’s build reproduces that paper’s local-proper-time-without-a-universal-tick signature directly. *Interference Gravity* needs the MOND interference term to be constructive; that constructive sign and V5’s attractive sign are the same coherence-content sign, supplied by *Phase Coherence in ED*’s operationalization of P12. And the earlier collective-pulse study, which found that the substrate’s crude shared-density coupling does *not* synchronize independent chain rhythms, left one open lead: whether a genuine cross-chain phase coupling would synchronize where shared density did not. It does, and it is the P12 coherence coupling. The “does a star pulse together” question and the V5 sign question turn out to be one object.

7. Falsifiers

- If Paper_090’s gauge law were the *sum* phase $e^{i(\alpha_A + \alpha_B)}$ rather than the difference, the conjugated-moment identification would break and V5’s coupling would be the joint-amplitude product, not the coherence content. It is the difference (§4.3).
- If the sign-flipped build had synchronized, the local lock would be an artifact of the coupling machinery and not of P12’s sign. It did not; the flip destroys binding.
- If a substrate build showed the coherence coupling, under the like-chain reading, anti-binds or fails to give local-without-global synchronization, the derivation fails. It gives exactly local-without-global.
- Empirical falsifiers are inherited from the companion papers (the clock-binding and MOND-interference observational content); this paper adds none.

8. Provenance audit

Tiers: **D** derived (from stated inputs), **D*** derived-conditional, **P** posited, **I** inherited value.

Element	Tier	Basis
13 primitives	P	Paper_087
V5 existence as a kernel rule-type	P	P10 licenses, does not force (090 §1.1); as for V1
V5 retarded support	D	P11 commitment-irreversibility (090 §3.4, §4.6)
V5 gauge-covariant phase	D	P05 + P09 (090 §4.3)
V5 coupling = coherence content of the cross-chain superposition	D*	gauge law difference-phase forces the conjugated-pair moment; conditional on P-Bipartite-Mapping + like-chain reading (§3)
V5 coupling is attractive (synchronizing)	D*	P12’s positive coherence sign, build-verified with a passing sign-flip control (§4)
Finite reach gives local-without-universal binding	D	build result, and the finiteness is load-bearing (§4)
ℓ_{V5} , τ_{V5} values	I	090 §3.3
Envelope shape F_{V5}	open	separate thread

End of draft (The Sign of the Binder). Companion to Paper_090 (V5 kernel), *The Relational Tick*, *Phase Coherence in ED*, and *Interference Gravity*. The build is `v5_coherence_coupling_probe.py`; the scoping and full result ledger are in `V5_ForwardDerivation_Scoping.md`.